Binary Classification using Artificial Neural Network (ANN)

Wisconsin Breast Cancer Dataset Project Explanation

Project Overview

In this project, we are building an Artificial Neural Network (ANN) model to perform **Binary Classification** on the Wisconsin Breast Cancer Dataset.

The objective of the project is to predict whether a breast tumor is:

- **Benign (Non-Cancerous)** → Safe
- **Malignant (Cancerous)** → Dangerous

using various medical measurements extracted from breast tissue samples.

Problem Statement

Breast cancer is one of the most common cancers affecting women worldwide. Early diagnosis plays a crucial role in increasing survival rates and reducing treatment costs.

Medical experts collect several measurements from breast cell images obtained through biopsy procedures. Based on these measurements, doctors determine whether the tumor is cancerous or non-cancerous.

Our goal is to train an Artificial Neural Network that can automatically learn patterns from historical patient data and assist in predicting tumor types.

Why is this a Binary Classification Problem?

In Machine Learning, classification problems are categorized based on the number of target classes.

Binary Classification

When the target variable contains only two classes, the problem is called Binary Classification.

In our dataset:

Target Value	Tumor Type
0	Malignant
1	Benign

Since there are only two possible outcomes, this becomes a Binary Classification problem.

Examples of Binary Classification:

- Spam or Not Spam
- Fraud or Genuine Transaction
- Disease or No Disease
- Pass or Fail
- Churn or Not Churn
- Malignant or Benign Tumor

Understanding the Dataset

The Wisconsin Breast Cancer Dataset contains:

- Total Samples: 569
- Total Features: 30
- Target Classes: 2

Each row represents one patient.

Each column represents a measurement extracted from a breast cell image.

Some important features include:

- Mean Radius
- Mean Texture
- Mean Perimeter
- Mean Area
- Mean Smoothness
- Mean Compactness
- Mean Concavity
- Mean Symmetry

These measurements describe the physical characteristics of a tumor.

Input and Output of the Model

Input

The ANN receives 30 numerical features as input.

Example:

Patient A:

- Radius = 17.99
- Texture = 10.38
- Area = 1001
- Smoothness = 0.1184
- Compactness = 0.2776

and many more features.

These values become the input neurons of the ANN.

Output

The ANN produces a single output value between 0 and 1 using the Sigmoid Activation Function.

Example:

Output = 0.95

This indicates a high probability of belonging to Class 1.

Prediction:

Benign

Another example:

Output = 0.08

This indicates a low probability of belonging to Class 1.

Prediction:

Malignant

Why Do We Use Sigmoid in the Output Layer?

The Sigmoid Activation Function converts any numerical value into a probability value between 0 and 1.

Formula:

Sigmoid(x) = 1 / (1 + e^(-x))

Output Range:

0 to 1

This makes Sigmoid suitable for Binary Classification problems.

Decision Rule

After obtaining the probability from the output neuron:

If Probability > 0.5

Prediction = Benign

If Probability ≤ 0.5

Prediction = Malignant

Example:

Probability	Prediction
0.90	Benign
0.75	Benign
0.30	Malignant
0.10	Malignant

ANN Architecture Used

The ANN consists of:

Input Layer

30 neurons

Each neuron receives one feature from the dataset.

Hidden Layer 1

16 neurons

Learns intermediate patterns from the input features.

Hidden Layer 2

8 neurons

Learns more complex relationships among features.

Output Layer

1 neuron

Uses Sigmoid Activation Function to predict probability.

Architecture:

Input Layer (30 Neurons)

↓

Hidden Layer 1 (16 Neurons)

↓

Hidden Layer 2 (8 Neurons)

↓

Output Layer (1 Neuron)

↓

Probability

↓

Benign or Malignant

Training Process

During training:

Step 1

Input data is passed through the network.

Step 2

The network generates predictions.

Step 3

Predictions are compared with actual labels.

Step 4

Error (Loss) is calculated using Binary Crossentropy.

Step 5

Backpropagation updates weights and biases.

Step 6

The process repeats for multiple epochs.

Gradually, the model learns patterns and improves its accuracy.

Why Feature Scaling is Important?

The dataset contains features with different ranges.

Example:

Area may have values around:

1000

while Smoothness may have values around:

0.1

Without scaling, larger values dominate the learning process.

Therefore, StandardScaler is applied before training the ANN.

This ensures all features contribute equally.

Model Evaluation

After training, the model is tested on unseen data.

Evaluation metrics include:

Accuracy

Measures overall correctness.

Formula:

Accuracy = Correct Predictions / Total Predictions

Confusion Matrix

Shows:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

Helps understand model behavior in detail.

Classification Report

Provides:

- Precision
- Recall
- F1-Score

for each class.

ROC Curve

Shows the trade-off between:

- True Positive Rate
- False Positive Rate

A larger area under the curve indicates better performance.

Real-World Impact

A trained ANN model can assist healthcare professionals by:

- Reducing manual effort
- Supporting diagnosis
- Identifying high-risk patients
- Improving decision-making

However, such models are not replacements for doctors.

They are decision-support systems that help medical experts make faster and more informed decisions.

Conclusion

In this project, we used an Artificial Neural Network to classify breast tumors as Benign or Malignant using 30 medical features from the Wisconsin Breast Cancer Dataset.

The ANN learns hidden patterns within the data through multiple layers of neurons and predicts the probability of a tumor being cancerous or non-cancerous.

This project demonstrates how Deep Learning can be applied to real-world healthcare problems and serves as a practical example of Binary Classification using Artificial Neural Networks.